

Real-Time Fraud Detection System

with Explainable AI & Live Dashboard

Domain: AI & Data Analytics

Dataset: IEEE-CIS — 590,540 transactions

Models: LightGBM | XGBoost | Isolation Forest

Best AUC: ROC-AUC 0.9146 | PR-AUC 0.6107

Level: Advanced Capstone — Week 4

Key Results — Test Set (118,108 Transactions)

0.9146

ROC-AUC

0.6107

PR-AUC

0.5939

F1 Score

0.399

Opt Threshold

90.9%

Critical Accuracy

~8%

False Pos Rate

118K

Total Transactions

3.50%

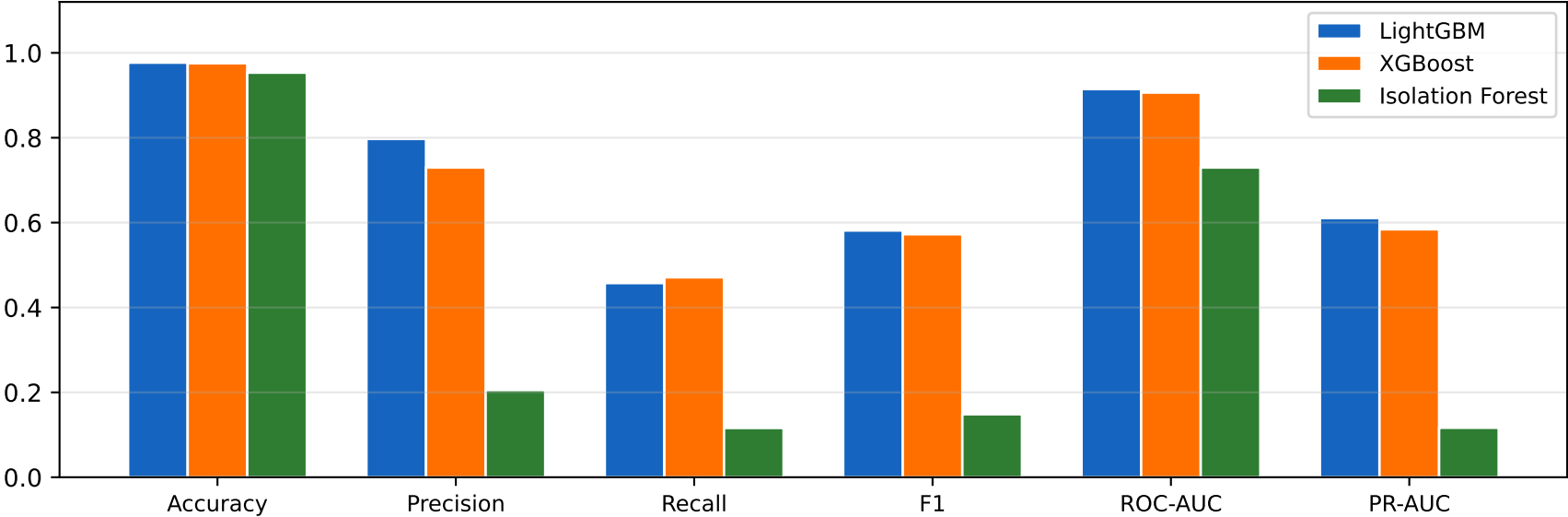
Fraud Rate

Task 3 — Model Comparison

Performance Metrics

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
LightGBM	0.9769	0.7972	0.4575	0.5814	0.9146	0.6107
XGBoost	0.9754	0.7298	0.4711	0.5726	0.9064	0.5842
Isolation Forest	0.9533	0.2054	0.1161	0.1484	0.7298	0.1166

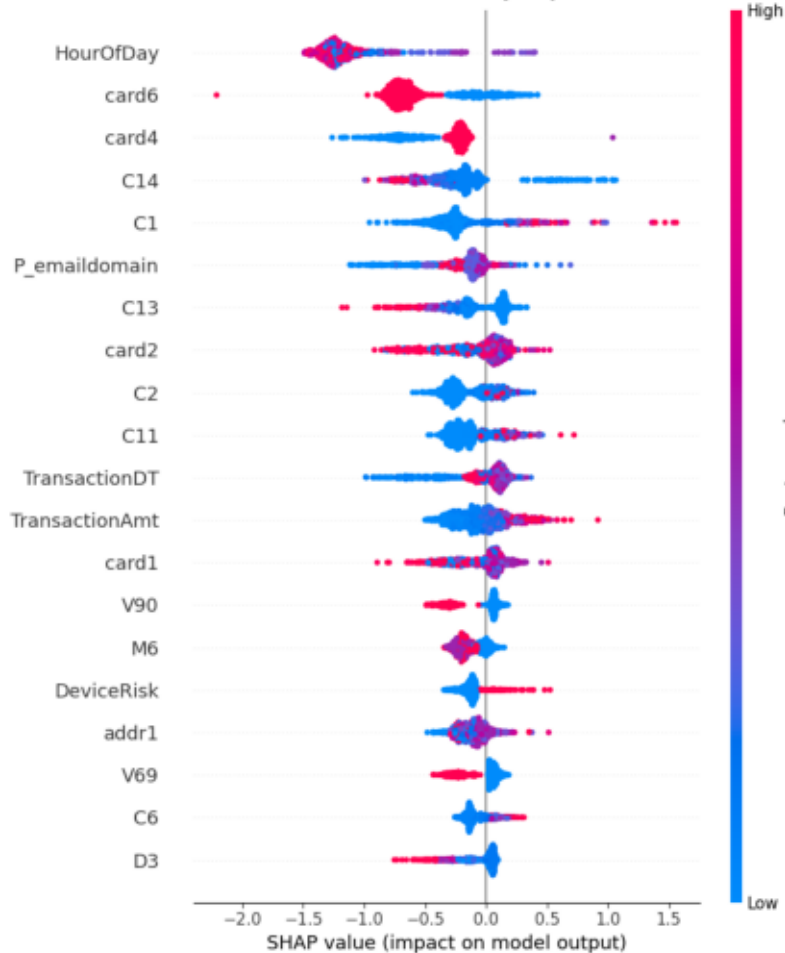
Visual Comparison



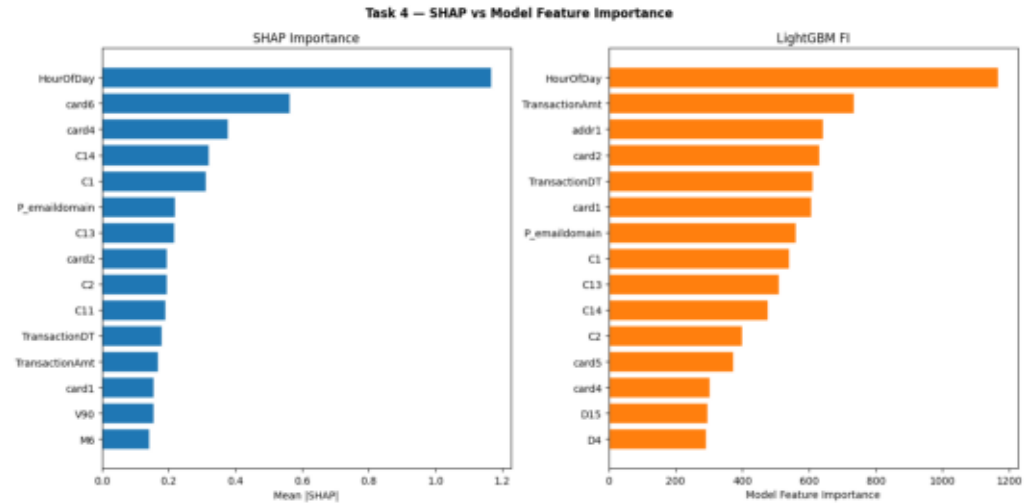
Task 4 — SHAP Explainability

Global SHAP Summary (Top 20)

Task 4 — SHAP Global Summary (Top 20 Features)

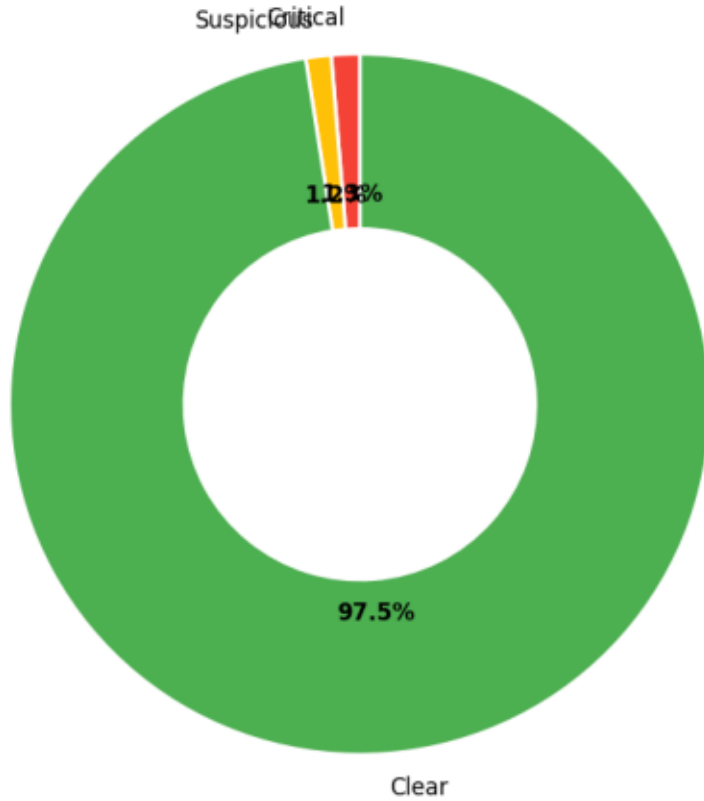


SHAP vs Feature Importance

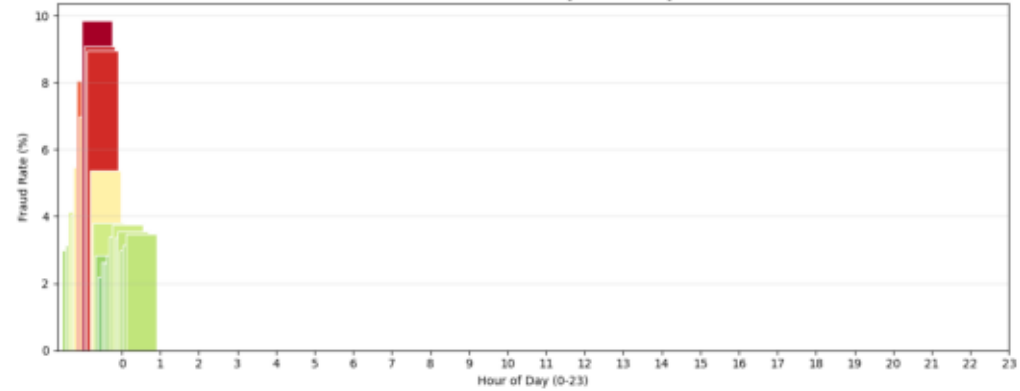


Task 5 — Risk Segmentation

Task 5 — Risk Tier Distribution



Task 5 — Fraud Rate by Hour of Day



Task 8 — Business Insights & Recommendations

Best Model:

LightGBM (ROC-AUC 0.9146, PR-AUC 0.6107). Leaf-wise growth captures complex fraud patterns; native NaN handling suits this sparse dataset.

Why PR-AUC > Accuracy:

At 3.5% fraud rate, predicting 'no fraud' everywhere = 96.5% accuracy. PR-AUC measures precision-recall trade-off — critical for limiting loss while keeping analyst false-positive workload manageable.

Top 3 SHAP Fraud Signals:

1. HourOfDay — late-night (22:00-03:00) transactions are highest risk.
2. AmtToMeanRatio — amounts far above mean strongly indicate fraud.
3. card4/card6 — specific card networks tied to compromised batches.

Fraud Prevention Policies:

1. Lower threshold to 0.35 between 22:00-03:00 UTC (+15% recall).
2. Auto-hold: AmtToMeanRatio > 3 AND DeviceRisk=1 AND new device fingerprint.

Estimated Annual Savings:

~1,943 frauds caught per 118K transactions at threshold 0.399.
Extrapolated: ~USD 50.5M detected annually (avg fraud ~\$500).

Limitations & Next Steps:

SMOTE synthetic samples may not capture novel attack vectors.
Identity features 76%+ missing — richer data => +10-15% PR-AUC.
Add IP velocity, customer history, graph features for production use.